AdeptiaFileTransferUtility

Version 1.0.0

Adeptia Custom Plugin — FTP / FTPS / SFTP File Routing

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1. Overview

AdeptiaFileTransferUtility is a custom Java plugin for the Adeptia integration platform that automates file transfer between FTP, FTPS, and SFTP servers with intelligent routing. It extracts a Group ID from each source filename, looks it up in a configurable XML mapping table, and delivers the file to the correct sub-directory on the target server — automatically.

Key Capabilities

- Supports FTP, Explicit FTPS, Implicit FTPS, and SFTP
- Extracts Group ID from filename pattern: prefix_GroupId_suffix.ext
- Maps Group ID to target sub-directory via SIC_EENAV.xml routing table
- Sequential or parallel transfer mode with configurable concurrency
- Auto-creates target directories if they do not exist
- Handles file conflicts by appending a timestamp to the filename
- Supports Adeptia Key Manager for SSH private key authentication
- Move mode (delete source after transfer) or Copy mode (keep source)

How It Works

The plugin performs the following steps for each file found in the source directory:

1. Connect to the source server using the configured FTP Account activity
2. List all files in the configured source directory
3. For each file: extract the Group ID from the filename
4. Look up the Group ID in SIC_EENAV.xml to find the target sub-directory (TransactionPath)
5. Connect to the target server using the configured FTP Account activity
6. Create the target sub-directory if it does not exist
7. Transfer the file to targetBasePath/TransactionPath/filename
8. Optionally delete the source file after successful transfer
9. Report results to the Adeptia context for downstream process flow use

> *Example: File prefix_101_Transaction.csv is transferred to /target/EENav_Custom1/prefix_101_Transaction.csv when Group ID 101 maps to TransactionPath EENav_Custom1.*

2. Prerequisites

Adeptia Platform

- Adeptia Connect 4.x or later
- Sys Admin access required to deployJar into Shared/Ext/

Adeptia Activities Required

- FTP Account activity for the source server
- FTP Account activity for the target server
- (Optional) Key Manager activity — required for SFTP with private key authentication
- Custom Plugin which invoke the Jar
- Multi Value Map Activity to move files as per GroupID

3. Implemention Steps

Step 1 — Copy the JAR File

Deploy the plugin JAR to the Shared/Ext directory. Refer to the below mentioned document for the same.

https://docs.adeptia.com/articles/#!ac47/custom-libraries/a/h2_755446135

Step 2 — Restart the Adeptia Microservices

Once the jar uploaded you need to Restart the Runtime and Webrunner Microservices to make the jar in affect.

Step 3 — Create a Multi Valued Map

Refer to the below document for creating Value Map.
https://docs.adeptia.com/articles/#!ac47/value-map/a/h2__1397387549

Once the value created value Map it will create an xml file on the below-mentioned location which is required in our custom plugin.
/shared/web/applets/mapping/multiValuedMap/global/<Multi Value Map Name>.xml

Getting the Activity ID

To get the FTP account entity Id, you can view the FTP account and copy the Entity ID.

View FTP: Test_Dummy_New

Description	Test_Dummy_New
Owner's Group	administrators
Entity Id	1485626149170913280
Type Id	FTPAccount:1485626149170913280
Creation Date	03/23/2026 18:37:56
Last Modified Date	03/23/2026 18:56:39
Last Modified By	admin
Permissions	Owner(R,W,X) Group(R,W,X)
Host Name	ftp2.adeptia.com
Port	22
FTP Type	SFTP
User Id	devteam
Password	**********
Transfer Type	PASSIVE
Transfer Mode	Binary

6. Get Target GroupId from Source File Name

The routing table maps a Group ID (extracted from the filename) to a target sub-directory (TransactionPath). It is a standard Adeptia MultiValueMap XML file.

Filename Pattern

The plugin extracts the Group ID from the second underscore-delimited segment of the filename:

```
Pattern:  prefix_GroupId_suffix.ext
Example:  EENav_101_Transaction.csv  ->  Group ID = "101"
Example:  SIC_202_Report.txt         ->  Group ID = "202"
```

Result of the Above Mapping

Source Filename	Group ID	Target Path
EENav_101_Txn.csv	101	/target/EENav_Custom1/EENav_101_Txn.csv
EENav_102_Txn.csv	102	/target/EENav_Custom2/EENav_102_Txn.csv
EENav_999_Unknown.csv	999	/target/EENav_Base/EENav_999_Unknown.csv

7. BeanShell Script — Use in the Custom Plugin

The Custom plugin is invoked from a BeanShell Script Activity inside your Adeptia process flow. Copy the script below, fill in your Activity IDs and paths, and paste it into the Script field of a BeanShell Script activity.

Parameter	Type	Default	Description
SrcAccountId	String	(required)	FTP Account Activity ID for the source server
TgtAccountId	String	(required)	FTP Account Activity ID for the target server
SourcePath	String	(required)	Full path to the source directory on the source server
TargetBasePath	String	(required)	Base path on the target server. Sub-directories are appended from the routing table
XmlPath	String	(required)	Absolute path to Multi Value Map xml file in th Adeptia Environment
parallelMode	boolean	false	false = process files one at a time. true = process files concurrently
concurrencyLimit	int	5	Maximum number of concurrent file transfers (applies only when parallelMode = true)
deleteFileOnSuccess	boolean	true	true = delete source file after successful transfer (Move mode). false = keep source file (Copy mode)

Context Variables Set by the Plugin

Context Key	Description
TotalFileReceived	Total number of files found in the source directory
TotalSuccessFiles	Number of files transferred successfully
TotalFailedFiles	Number of files that failed to transfer
FailedFileList	List of files which fail to process
CurrentFTPActiveConnection	Set only if connections were not properly closed — indicates an error condition

9. Process Flow Setup in Adeptia

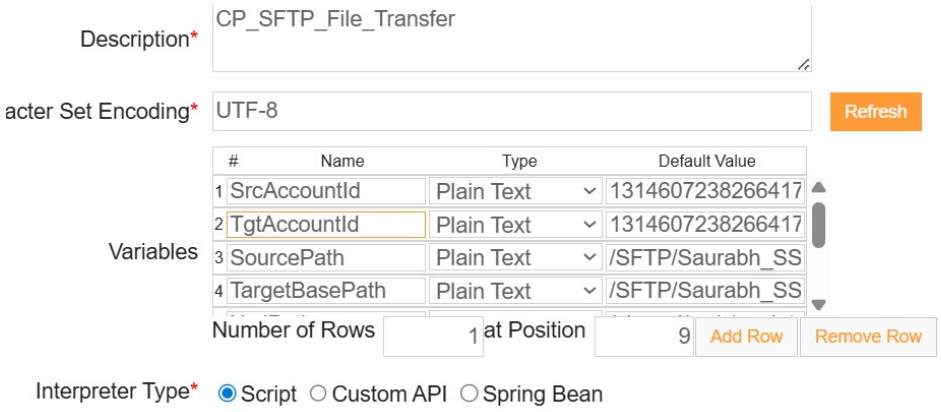
Follow these steps to set up a complete process flow that uses the plugin.

Step 1 — Create a New Process Flow

10. In Adeptia Designer go to Services > Process Flow > Create New
11. Give the process flow a descriptive name (e.g. SIC_FTP_FileRouter)

Step 2 — Add a Custom Plugin Activity

12. Create a custom plugin activity and use the attached custom plugin in the CP_FTP_FileTransfer.txt file.
13. Update SrcAccountId, TgtAccountId, SourcePath, TargetBasePath, and XmlPath, ParallelMode, concurrencyLimit with your actual values. Refer to the attached screenshot.



14. Save the activity

10. Sequential vs Parallel Mode

	Sequential (parallelMode=false)	Parallel (parallelMode=true)
Connections	1 source + 1 target connection shared across all files	1 source + 1 target connection per concurrent file
Speed	Slower — files processed one at a time	Faster — multiple files processed simultaneously
Server load	Low — single connection	Higher — multiple concurrent connections
Best for	Servers with connection limits, small batches	Large batches, servers that allow concurrent connections
Connection count	2 (1 src + 1 tgt)	2 x concurrencyLimit (up to 2 per thread)